VOICE OPERATION NOT POSSIBLE [ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY]]

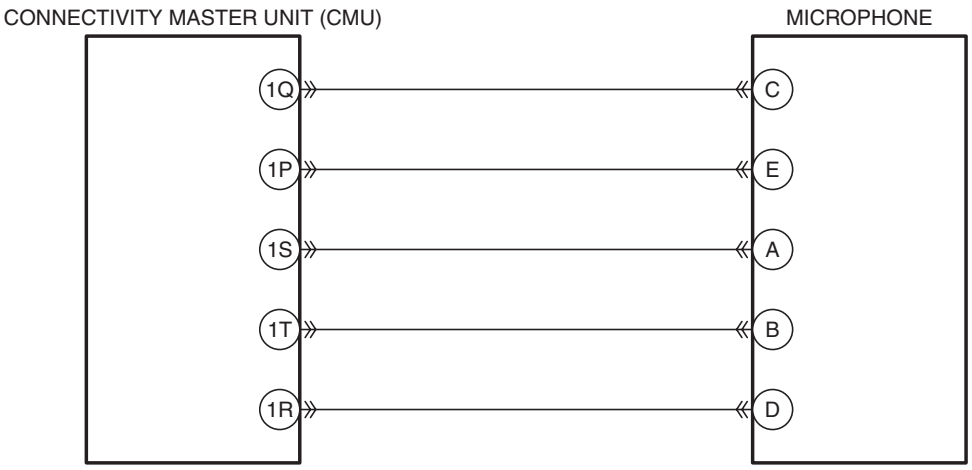
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Description	**VOICE OPERATION NOT POSSIBLE** Voice operation of audio function, communication function, and navigation function is not possible **Note** • The following applicable voice operations are not possible. — Voice operation of audio function • Except for USB audio mode, voice operation for specifying, song title, artist name, and album name is not possible • If the USB device audio file does not contain song information (song title, artist name, album name), the song information cannot be specified. — Hands-free telephone (HF/TEL) voice operation • If the connectivity master unit (CMU) cannot read the telephone book, voice operation is not possible. • If “Mr./Mrs.” is added to a person's name when speaking, normal voice operation is not possible (non-match to name registered in telephone book) • If there are several names with similar pronunciation to the name registered in the telephone book, normal voice operation is not possible (John Smith/ Juan Smith) • If the person's name that is spoken is short, normal voice operation is not possible (Ma/Pa)
Possible Causes	• Malfunction due to conditions of use — Abnormally slow speech — Loud speech — Difference between language set in the CMU and language used in voice operation • Malfunction due to effects of surrounding environment — Air from A/C is blown against the microphone — Loud road noise, A/C operation sound, external noise • Speaker malfunction • Tuner and amp unit (TAU) malfunction • Connector poor contact or terminal damage • Microphone malfunction — Open circuit in wiring harness between the following terminals • CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) • CMU terminal 1R and microphone terminal D • CMU terminal 1S and microphone terminal A • CMU terminal 1T and microphone terminal B — Short to ground in wiring harness between the following terminals • CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) — Short to power supply in wiring harness between the following terminals • CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) • CMU terminal 1R and microphone terminal D — Microphone internal malfunction • CMU internal malfunction

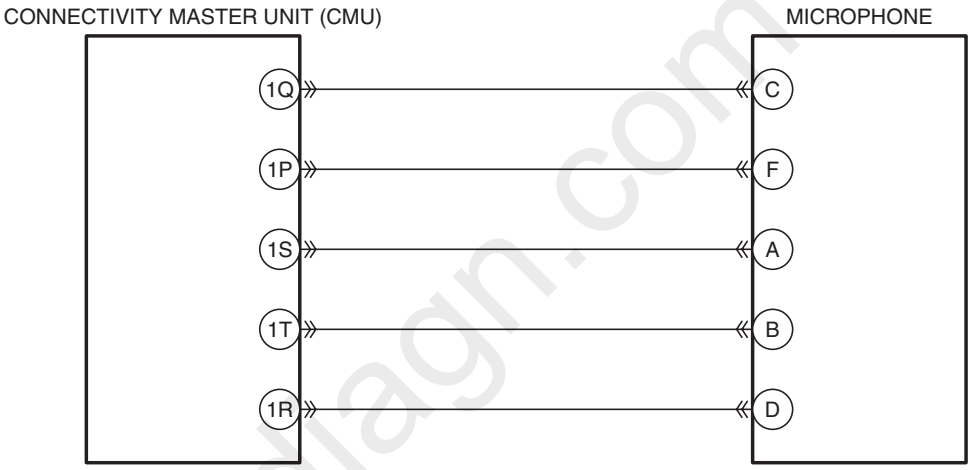
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System Wiring Diagram

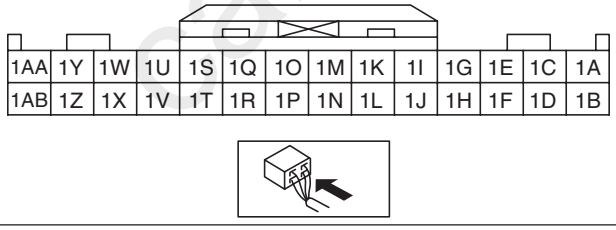
L.H.D.



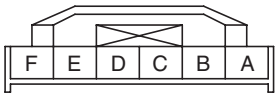
R.H.D.



CONNECTIVITY MASTER UNIT (CMU)
WIRING HARNESS-SIDE CONNECTOR



MICROPHONE
WIRING HARNESS-SIDE CONNECTOR



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Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY IF MALFUNCTION DUE TO CUSTOMER USE METHOD** • Switch the ignition to ACC or ON (engine off or on). • Press the TALK button on the steering switch. • After the voice guidance instructs the user to speak after the beep, say, “AM” into the microphone in a normal or slightly loud voice. • Does the center display change to the AM radio screen? **Determination criteria** — Screen changes to AM screen once or more by voicing 10 times.	Yes	System is normal. (Explain the contents of the servicing to the customer)
		No	Go to the next step.

Step	Inspection		Action
2	**VERIFY IF MALFUNCTION DUE TO EFFECTS OF SURROUNDING ENVIRONMENT** • Verify if air from the A/C is being blown against the microphone. • Move the vehicle to a location with little external noise. • Press the TALK button on the steering switch. • After the voice guidance instructs the user to speak after the beep, say, “AM” into the microphone in a normal or slightly loud voice. • Does the center display change to the AM radio screen? **Determination criteria** — Screen changes to AM screen once or more by voicing 10 times.	Yes	System is normal. (Explain the contents of the servicing to the customer)
		No	Go to the next step.
3	**DETERMINE IF MALFUNCTION IS SPEAKER, AUDIO AMPLIFIER RELATED** • Switch the ignition to ACC or ON (engine off or on). • Launch the on-board diagnostic assist function. (See DIAGNOSTIC ASSIST FUNCTION [CONNECTIVITY MASTER UNIT].) • Select assist code “94”. • Press the [ENTER] and verify that the sound is output from each speaker. • Is sound output from each speaker?	Yes	The speakers and audio amplifier are normal. Go to the next step.
		No	Perform malfunction diagnosis according to NO SOUND OUTPUT IN ALL MODES. (See NO SOUND OUTPUT IN ALL MODES [ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY]].)
4	**DETERMINE IF MALFUNCTION IS TAU** • Switch the ignition to ACC or ON (engine off or on). • Operate the command switch or center display and verify if sound is output from the following modes. — Bluetooth® audio — Internet radio — USB audio • Is sound output?	Yes	TAU is normal. Go to the next step.
		No	Perform malfunction diagnosis according to NO SOUND OUTPUT IN MULTIPLE MODES. (See NO SOUND OUTPUT IN MULTIPLE MODES [ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY]].)
5	**DETERMINE IF MALFUNCTION IS MICROPHONE RELATED** • Switch the ignition to ACC or ON (engine off or on). • Launch the on-board diagnostic assist function. (See DIAGNOSTIC ASSIST FUNCTION [CONNECTIVITY MASTER UNIT].) • Select assist code “61”. • Press ENTER. • Say, “AM” into the microphone in a normal or slightly loud voice. • When speaking, verify if the “Microphone” item is displayed. • Is the “Microphone” item displayed?	Yes	The microphone is normal. Go to Step 10.
		No	Go to the next step.
6	**MICROPHONE CONNECTOR INSPECTION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the microphone connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 11.

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Step	Inspection		Action
7	**VERIFY IF MALFUNCTION CAUSE IS OPEN CIRCUIT IN WIRING HARNESS BETWEEN CMU AND MICROPHONE** • Switch the ignition off. • Disconnect the CMU connector. • Verify that the microphone connector is disconnected. • Inspect the wiring harness for continuity between the following terminals (vehicle wiring harness side). — CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) — CMU terminal 1R and microphone terminal D — CMU terminal 1S and microphone terminal A — CMU terminal 1T and microphone terminal B • Is there continuity?	Yes	Go to the next step.
		No	• Refer to the wiring diagram and verify if there is a common connector between the following terminals. — CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) — CMU terminal 1R and microphone terminal D — CMU terminal 1S and microphone terminal A — CMU terminal 1T and microphone terminal B **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. • Go to Step 11.
8	**VERIFY IF MALFUNCTION CAUSE IS SHORT CIRCUIT TO GROUND IN WIRING HARNESS BETWEEN CMU AND MICROPHONE** • Verify that the CMU and microphone connectors are disconnected. • Inspect for continuity between the following wiring harness terminals (vehicle wiring harness side) and body ground. — Microphone terminal E (L.H.D.)/F (R.H.D.) • Is there continuity?	Yes	• Refer to the wiring diagram and verify if there is a common connector between the following terminals. — CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Go to Step 11.
		No	Go to the next step.
9	**VERIFY IF MALFUNCTION CAUSE IS SHORT CIRCUIT TO POWER SUPPLY IN WIRING HARNESS BETWEEN CMU AND MICROPHONE** • Verify that the CMU and microphone connectors are disconnected. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition to ACC or ON (engine off or on). • Measure the voltage at the following terminals (vehicle wiring harness side). — Microphone terminal E (L.H.D.)/F (R.H.D.) — Microphone terminal D • Is the voltage **B+**?	Yes	• Refer to the wiring diagram and verify if there is a common connector between the following terminals. — CMU terminal 1P and microphone terminal E (L.H.D.)/F (R.H.D.) — CMU terminal 1R and microphone terminal D **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to power supply to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to the power supply. • Go to Step 11.
		No	Replace the microphone, then go to Step 11. (See MICROPHONE REMOVAL/INSTALLATION.)
10	**INSPECT CMU CONNECTOR** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the CMU connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Replace the CMU, then go to the next step. (See CONNECTIVITY MASTER UNIT (CMU) REMOVAL/INSTALLATION.)
		No	Repair or replace the connector, then go to the next step.

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Step	Inspection		Action
11	**VERIFY IF MALFUNCTION CAUSE IS CORRECTED** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect all the connectors. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition to ACC or ON (engine off or on). • Move the vehicle to a location with little external noise. • Press the TALK button on the steering switch. • After the voice guidance instructs the user to speak after the beep, say, “AM” into the microphone in a normal or slightly loud voice. • Does the center display change to the AM radio screen? **Determination criteria** — Screen changes to AM screen once or more by voicing 10 times.	Yes	Troubleshooting completed. (Explain the contents of the servicing to the customer.)
		No	Verify the malfunction symptom in the symptom troubleshooting chart and perform the other applicable malfunction diagnosis. (See SYMPTOM TROUBLESHOOTING [ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY]].)

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